



On Solving the Network-Based Portfolio Optimization Problem

Igor Cardoso (Igor.Cardoso@ttu.edu), Hamidreza Validi (hvalidi@ttu.edu)

Department of Industrial, Manufacturing & Systems Engineering, Texas Tech University



Motivation & Contributions

- **Context:** Modern portfolio optimization balances expected return and risk while integrating structural information from financial networks.
- **Challenge:** Enforcing a star diversification constraint directly on the market correlation graph embeds a complex Maximum Independent Set problem, making exact optimization intractable.
- **Contributions:**
 - 1 Shortfall MIP reformulation with tight Big- M bounds
 - 2 Five families of valid inequalities (e.g., maximal-clique, odd-hole)
 - 3 Proof of strong NP-hardness and inapproximability
 - 4 A Greedy Independent-Set Portfolio (GISP) heuristic

Problem Statement

Problem: Network-Based Portfolio Optimization (NBPO).

Input: A graph $G_\theta = (V, E)$ with threshold θ , returns r_{it} , expected returns μ_i , minimum weight γ , and violation tolerance $\delta \in (0, 1)$.

Output: An allocation x maximizing $\sum_i \mu_i x_i$ subject to a chance-return constraint and star diversification in G_θ .

Notations

r_{it}, μ_i	Scenario return and expected return of asset i on day t
$G_\theta = (V, E)$	Market correlation graph (edge $\Leftrightarrow$ correlation $\geq \theta$)
$N_{G_\theta}[i]$	Closed neighborhood of asset i in G_θ
T	Set of historical trading days (scenarios); $t \in T$
R^{VAR}	Minimum acceptable daily portfolio return
M_t	Tight, cardinality-valid Big- M bound for shortfall s_t
δ	Maximum allowed fraction of days violating R^{VAR}
γ	Minimum portfolio weight for any selected asset
x_i, y_i, z_t, s_t	Weight, selection, violation, and shortfall variables

MIP Shortfall Formulation for NBPO

$$\begin{aligned}
 \max \quad & \sum_{i \in V} \mu_i x_i \\
 \text{s.t.} \quad & \sum_{i \in V} r_{it} x_i + s_t \geq R^{\text{VAR}} \quad \forall t \in T \quad (\text{shortfall}) \\
 & s_t \leq M_t z_t \quad \forall t \in T \quad (\text{shortfall link}) \\
 & \frac{1}{|T|} \sum_{t \in T} z_t \leq \delta \quad (\text{violation budget}) \\
 & \sum_{i \in V} x_i = 1 \quad (\text{full invest.}) \\
 & \sum_{j \in N_{G_\theta}[i]} y_j \leq 1 \quad \forall i \in V \quad (\text{star div.}) \\
 & \gamma y_i \leq x_i \leq y_i \quad \forall i \in V \quad (\text{weight bounds}) \\
 & y_i, z_t \in \{0, 1\} \quad \forall i \in V, t \in T \\
 & x_i, s_t \geq 0 \quad \forall i \in V, t \in T.
 \end{aligned}$$

Advantage of the Shortfall Reformulation

The standard formulation enforces the chance constraint directly within the return equation. By separating the return-deficit accounting (s_t) from the binary activation (z_t), our reformulation prevents the LP from using fractional z_t values to artificially boost returns on bad days.

Standard Constraint:

$$\sum_{i \in V} r_{it} x_i + M_t z_t \geq R^{\text{VAR}}$$

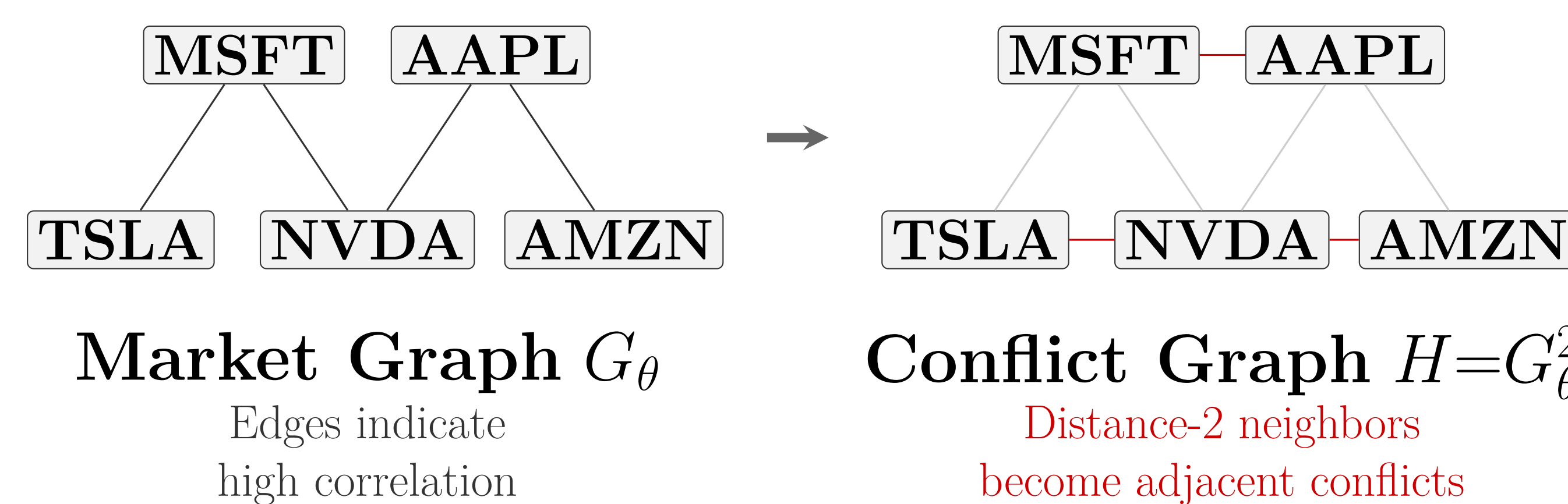
Shortfall Constraints:

$$\sum_{i \in V} r_{it} x_i + s_t \geq R^{\text{VAR}}$$

$$s_t \leq M_t z_t$$

Preliminary Results: This decoupling exposes explicit s_t variables required for valid inequalities and stops fractional LP artifacts. **Empirical Impact:** The shortfall LP is structurally tighter (firing $\sim 45\%$ fewer clique cuts). On hard instances, the shortfall MIP with cuts delivers a **2.5–3x B&B node-count reduction** versus the standard MIP.

Market Graph & Conflict Graph



Star diversification $\Leftrightarrow$ independent set in $H = G_\theta^2$.

Valid Inequalities for the NBPO Polytope

We derive five families of valid inequalities exploiting the independent-set structure on the conflict graph $H = G_\theta^2$ and the continuous shortfall variables:

- **Maximal-Clique:** For any clique $C \subseteq V$ in H , $\sum_{i \in C} y_i \leq 1$
- **Odd-Hole:** For an induced odd cycle $(v_1, \dots, v_{2p+1})$ in H , $\sum_{j=1}^{2p+1} y_{v_j} \leq p$
- **Cardinality-Allocation:** For each asset $i \in V$, $x_i + \gamma \sum_{j \in V \setminus \{i\}} y_j \leq 1$
- **Clique-Weight:** For any clique $C \subseteq V$ in H , $\sum_{i \in C} x_i + \gamma \sum_{j \in V \setminus C} y_j \leq 1$
- **Polymatroid (Mixing):** For subsets of days $J \subseteq Q \subseteq T$ ranked by M_t , $\sum_{t \in Q} s_t \leq \sum_{t \in J} M_t$

Preliminary Computational Impact of Valid Inequalities

Evaluated via Gurobi user-cut callbacks across 30 instances (120s limit). Clique cuts on H dominated all others, firing within the branch-and-bound tree:

Performance Metric	Result
Total Clique Cuts Added (30 Instances)	478,203
Node Reduction (On 21 Time-Limited Instances)	20 / 21 (95%)
Median B&B Node Reduction	61%
Median MIP-Gap Closure	72%

Takeaway: Clique cuts accelerate convergence on hard instances. For example, on an $n = 40$ scale-up instance, the node count dropped from 331,123 to just 83,486.

Strong NP-Hardness & Inapproximability

Theorem: NBPO is strongly NP-hard. Furthermore, via a subdivision graph construction reducing from Maximum Weight Independent Set, unless $P = NP$, no polynomial-time algorithm approximates NBPO within a factor of $|V|^{1/2-\epsilon}$.

The inapproximability transfers directly from the MIS-hardness of independent-set selection in the conflict graph $H = G_\theta^2$.

The GISP Heuristic

We propose the **Greedy Independent-Set Portfolio (GISP)** algorithm:

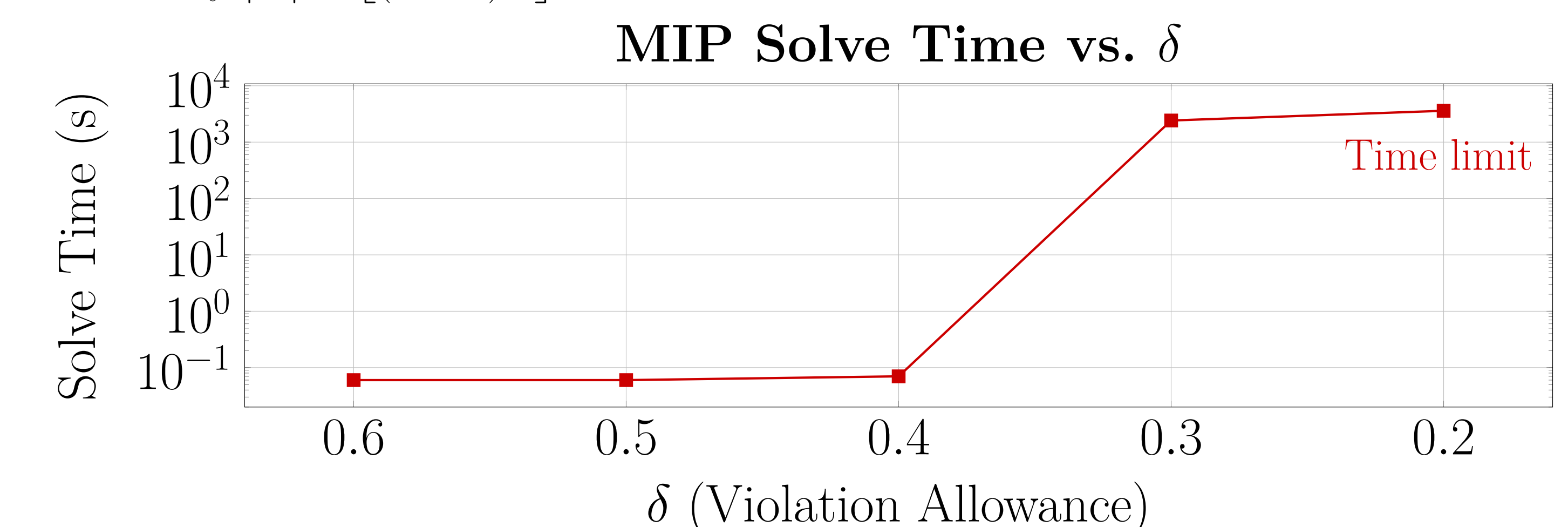
- 1 **Asset Ranking:** Sort assets by expected return μ_i (descending).
- 2 **Graph Selection:** Greedily add each asset if it introduces no conflict in H .
- 3 **Scenario Filtering:** Isolate worst-case historical returns up to the violation budget δ using static day pre-selection.
- 4 **Continuous Allocation:** Solve the residual allocation LP over the selected independent set and filtered days.

Complexity: $O(n \log n + n \Delta(G_\theta)^2 + T \log T)$.

Guarantee: $\frac{1}{\Delta(H)+1}$ -approximation on the selection component.

Computational Phase Transition

Experiments reveal a sharp transition in solver difficulty governed by the active-day cardinality $|\hat{A}| = \lfloor (1 - \delta)T \rfloor$:



- **Loose regime** ($\delta \geq 0.4$): NBPO solved optimally in under one second.
- **Tight regime** ($\delta \leq 0.3$): Solver hits a bottleneck; GISP produces high-quality feasible portfolios in < 0.01 s.

Conclusions & Future Work

- We establish strong theoretical foundations for NBPO: NP-hardness, an allocation sparsity theorem, valid inequalities, and a dominance reduction rule.
- The δ -phase transition identifies an inherent computational barrier separating tractable from hard instances.
- **Future work:** Branch-and-price decomposition, stronger LP relaxations via lift-and-project, and multi-period extensions.

References

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- [3] Ricca, F., Scozzari, A. Portfolio optimization through a network approach: Network assortative mixing and portfolio diversification. *European Journal of Operational Research*, 312(2):700–717, 2024.